S&P GLOBAL & CRISIL CAMPUS HACKATHON 2026

Real-Time AI Risk Signal Engine & Tactical Index Rebalancer

From unstructured news & social text to structured risk signals, and a sentiment-tilted index (Module A)

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PROBLEM & APPROACH

Too much text, too little structure

Market-moving information arrives as headlines and posts. Analysts can't read it all, and keyword alerts are noisy and not comparable across companies. Goal: turn the stream into explainable numbers a portfolio can act on.

1 Read

News: GDELT live API + Kaggle financial news

Social: Kaggle stock tweets

Synthetic sample so the demo always runs offline

Clean, dedupe, link to 15 S&P 100 tickers

2 Score

Sentiment -1..1 (FinBERT)

Event type 10 classes (rules)

Impact 1..10 (explainable formula)

One JSON signal per company per document

3 Act - Module A

Decayed, impact-weighted sentiment per stock

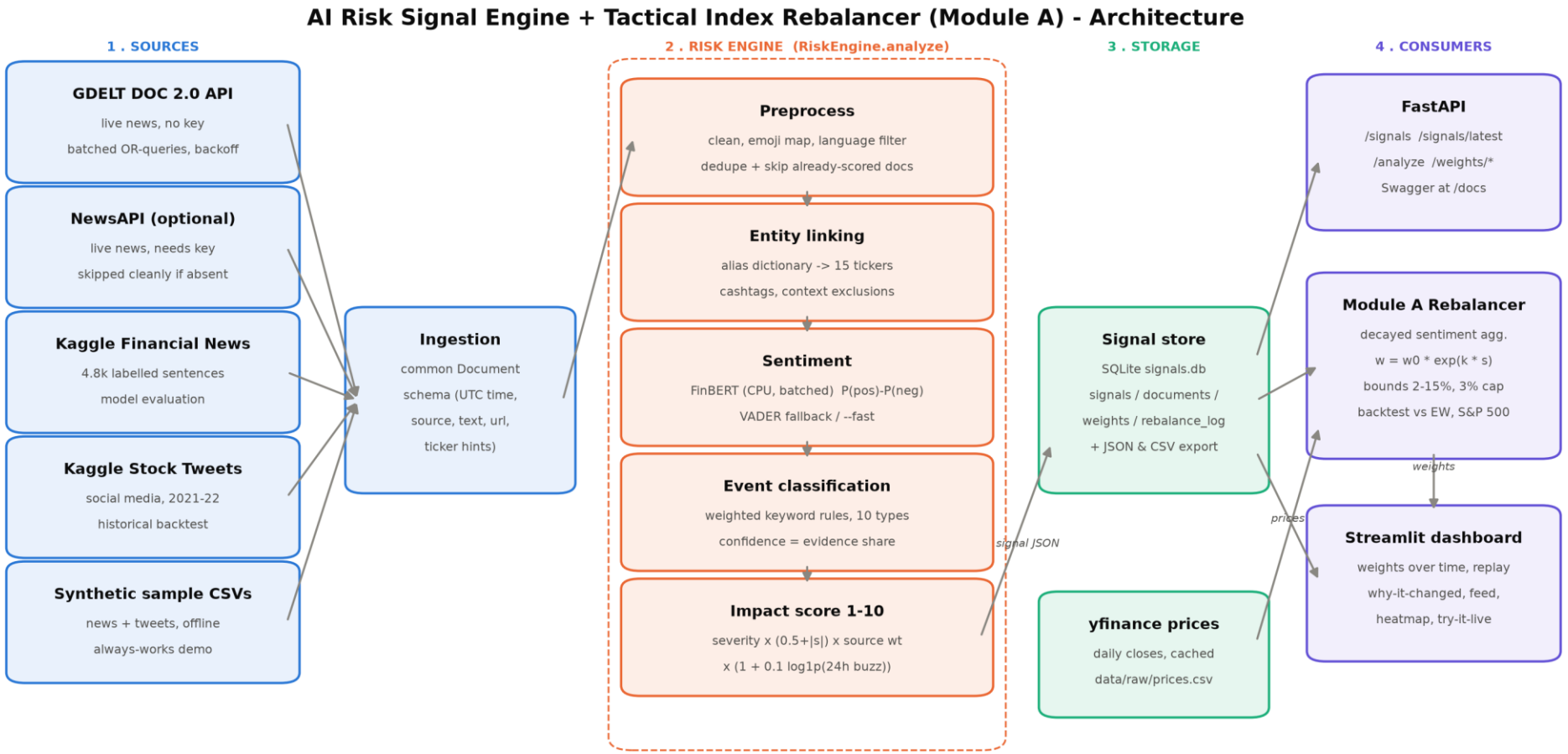
Tilt equal weights: w = w0 · e^(k·s)

Bounds 2-15%, max 3 pts change / day

Backtest vs equal weight & S&P 500

ARCHITECTURE

System design and data flow



No look-ahead: weights set from signals up to day t's close earn day t+1's return. Everything runs offline on cached data; live sources fall back gracefully.

KEY MODULES & WHY

Implementation highlights

FinBERT sentiment

Finance-tuned BERT on CPU, batched; score = P(pos) − P(neg). Cached so no text is scored twice. VADER fallback (--fast).

Rule-based events

Weighted regex per class (credit rating, takeover, antitrust, CEO resigns…). Auditable; confidence = share of evidence.

Explainable impact

severity[event] × (0.5+|s|) × source weight × (1+0.1·ln(1+24h buzz)), clipped 1-10. All weights in config.yaml.

Bounded rebalancer

Exact solver: bisection keeps Σw = 1 with 2-15% bounds and a 3-pt turnover cap. Each change logged with its headlines.

No look-ahead

Signals up to 4pm NY on day t set weights that earn day t+1's return. Enforced by a unit test (169 tests).

Serving

SQLite + JSON/CSV export · FastAPI (/signals, /analyze, /weights) · Streamlit dashboard · live GDELT polling.

OUTPUTS & METRICS

Key results

Sentiment model (labelled financial news)

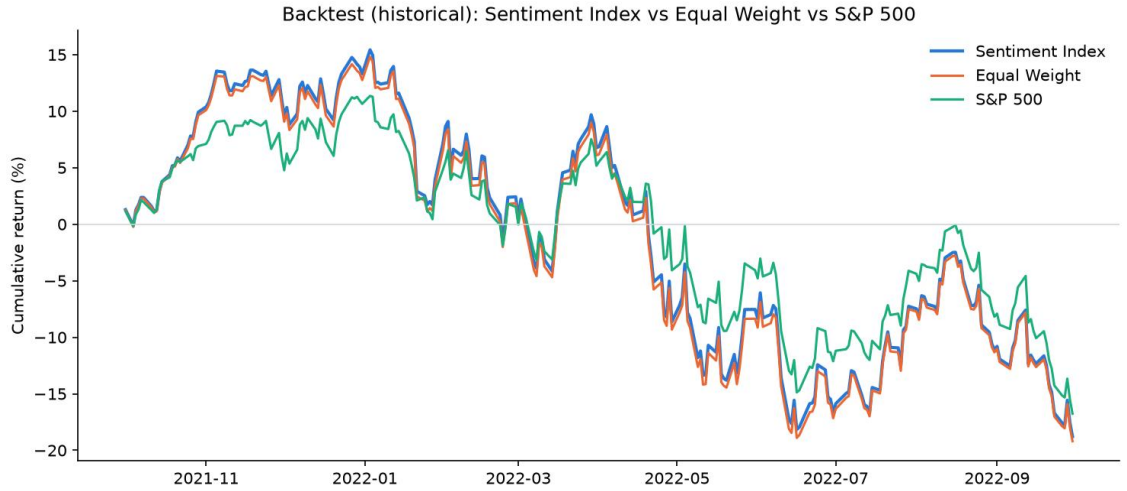
Model	Accuracy	Macro-F1
FinBERT	88.9%	0.882
VADER baseline	54.4%	0.489

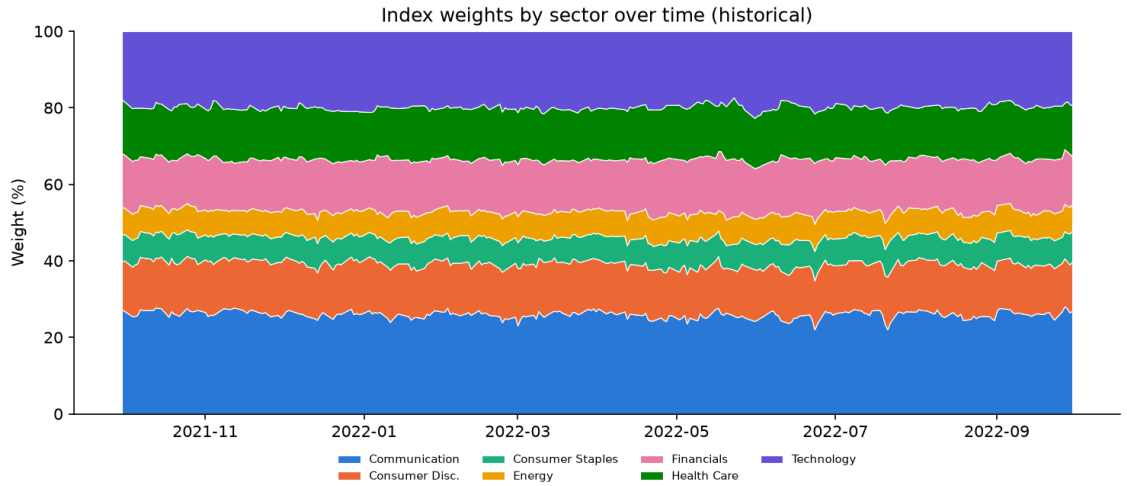
n = 4,838 sentences (Kaggle / Financial PhraseBank)

Backtest (Kaggle tweets, Sep 2021 – Sep 2022)

Strategy	Return	Sharpe	Max DD
Sentiment Index	-18.8%	-0.65	-29.6%
Equal Weight	-19.2%	-0.66	-29.6%
S&P 500	-16.8%	-0.73	-25.2%

Avg daily turnover 2.3%; no transaction costs. Illustrative only.





DOMAIN IMPACT

Why it matters

Faster reaction

A headline becomes a scored, company-tagged signal in well under a second on a laptop CPU, and live mode refreshes every 15 minutes.

Explainable by design

Class probabilities, matched event keywords, the impact formula and the headlines behind every weight change are all visible. That fits risk and index governance.

Controlled risk

Weight bounds, a turnover cap and decay to neutral stop the index from chasing one headline.

Reusable engine

The same API feed can drive watch-lists, credit early-warning, or stress-testing scenarios (Module B), not just this index.

499 signals from 3 sources currently in data/signals.json, all served by FastAPI and the dashboard.

HONEST SCOPE

Limitations & next steps

Limitations

One sentiment per document: two companies in a headline share it

Dictionary entity linking; ambiguous names need exclusion lists

Keyword event rules miss unusual phrasing (falls to “Other”)

FinBERT is trained on news; weaker on slang-heavy tweets

Backtest: one year of tweets, no transaction costs

GDELT throttling: a live poll takes 2-4 minutes

Next steps

Entity-level (aspect) sentiment

Fine-tune FinBERT on labelled finance tweets

Zero-shot classifier as a second opinion on events

Transaction-cost and liquidity-aware rebalancing

Longer, multi-source history (news archives)

Module B: feed signals into stress-testing scenarios